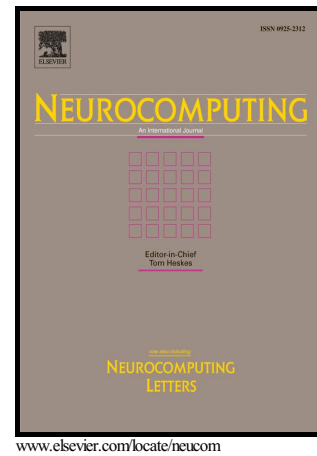


Author's Accepted Manuscript

Graph Coloring based Surveillance Video Synopsis

Yi He, Changxin Gao, Nong Sang, Zhiguo Qu, Jun Han



PII: S0925-2312(16)31340-6
DOI: <http://dx.doi.org/10.1016/j.neucom.2016.11.011>
Reference: NEUCOM17709

To appear in: *Neurocomputing*

Received date: 24 February 2016
Revised date: 21 September 2016
Accepted date: 9 November 2016

Cite this article as: Yi He, Changxin Gao, Nong Sang, Zhiguo Qu and Jun Han Graph Coloring based Surveillance Video Synopsis, *Neurocomputing* <http://dx.doi.org/10.1016/j.neucom.2016.11.011>

This is a PDF file of an unedited manuscript that has been accepted for publication. As a service to our customers we are providing this early version of the manuscript. The manuscript will undergo copyediting, typesetting, and review of the resulting galley proof before it is published in its final citable form. Please note that during the production process errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

Graph Coloring based Surveillance Video Synopsis[☆]

Yi He^a, Changxin Gao^{a,*}, Nong Sang^a, Zhiguo Qu^b, Jun Han^b

^a*School of Automation, 1037 Luoyu Road, Wuhan 430074, China*

^b*Air Force Early Warning Academy, 288 Huangpu Road, Wuhan 430019, China*

Abstract

Video synopsis is an intelligent condensation approach to solve fast video browsing and retrieval for surveillance cameras. However, collision caused by unsatisfied tube rearrangement in traditional methods brings uncomfortable visual effect to users and how to mitigate the collision still remains an attracting topic. Unlike conventional methods that deal with tube rearrangement by minimizing a global energy function, we propose a novel approach by formulating it as a graph coloring problem. In our approach, all the tubes are firstly mapped into the spatial domain for analyzing their potential collision relationship. The input tube set is then represented by a graph structure, where each node stands for a tube and the edge between two nodes represents the potential collision relationship. To mitigate the collision artifacts, our method finds the mapping of tubes from original video to synopsis video by $L(q)$ -coloring the graph, which separates tubes from their collision points. The parameter q is left tunable to make a compromise between collision artifacts and synopsis length, which can better meet users' demand of freely adjusting the compactness of synopsis video. The shifted objects are finally composited with the background image to obtain the high-quality video synopsis. Extensive experimental results show that the proposed method can generate more compact video synopsis with less collision artifacts than the existing methods.

Keywords: video synopsis, tube rearrangement, graph coloring

[☆]Fully documented templates are available in the elsarticle package on CTAN.

*Corresponding author

Email addresses: heyi2012@hust.edu.cn (Yi He), cgao@hust.edu.cn (Changxin Gao)

1. Introduction

With the rapid development of site-seeing cameras and ever-growing surveillance raw data, it is time consuming and labor wasting to review the entire videos and find interesting events from such huge data. Thus, constructing condensed representations of long videos to help users browse raw data quickly and efficiently become an urgent issue to be solved in the security industry. In addition, it is memory intensive to store the entire captured videos. Fortunately, video condensation can effectively reduce memory storage for large data sets. Various condensation methods have been proposed to generate meaningful representations of original videos, such as video fast forward [1, 2, 3], video skimming [4], video abstraction [5, 6, 7, 8], video summarization [9, 10, 11, 12, 13, 14], video montage [15, 16, 17], and video synopsis [18, 19, 20, 21]. These methods can be generally classified into two broad categories: frame-based approaches and object-based approaches.

The frame-based approaches, which take entire image frames as the basic building blocks, are earlier developed methods to obtain short video representations. The aforementioned methods such as video fast forward, video skimming, video abstraction and video summarization, *et al.* all belong to this category. Such methods either select several key frames according to certain important criteria or skip uninteresting frames in periods of low activity to condense the video for fast browsing. These methods lose the dynamic aspect of videos and may cause important contents missing. Comprehensive surveys on frame-based approaches are given in [6, 7, 8].

The object-based approaches, including video montage and video synopsis, are recently developed to generate more compact video representation, where objects from different times are used as the processing units. The space-time video montage [15], extracts informative spatial-temporal portions and packs them together, to create a condensed video. However, videos condensed in this way have perceptible information loss and obvious seams at the boundaries

30 between different portions. A significant step forward in the object-based approaches is video synopsis [18, 19, 20], in which a tube (frame sequences of an object) is considered as the basic processing unit. Video synopsis aims to produce a shortened video sequence by rearranging the tubes appropriately, which is shifting the tubes either in time domain, or in both domain. Compared with
 35 other methods, video synopsis can achieve a high condensation ratio, which may enable users to browse a day’s long surveillance video in just a few minutes. Meanwhile, all activities and dynamics of moving objects can be completely retained.

Based on the proposition of video synopsis framework, several issues with
 40 respect to condensation speed and tube rearrangement strategies have been addressed in [22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35]. Generally speaking, the pipeline of video synopsis includes background generation, object segmentation and tube extraction, tube rearrangement and synopsis composition, in which tube rearrangement plays a key role in the whole process
 45 to achieve a high condensation ratio. According to whether tube rearrangement is implemented online or not, current video synopsis methods can be further divided into online video synopsis [25, 26, 28, 31, 32, 33] and offline [22, 23, 24, 27, 29, 30, 34, 35] video synopsis. However, tube rearrangement in all these methods largely follows the same idea, which is formulated as
 50 a constrained optimization problem. By minimizing a global energy function, tube rearrangement determines the tubes’ start-time labels so that the tubes originally appearing in different times can be displayed simultaneously and thus condense the original videos. The global energy function to be minimized usually consists of several costs under some constraints, such as the maximum
 55 activity, background consistency, chronological order, tube collision, and so on. To minimize the energy function, a simple greedy optimization algorithm or graph cuts is usually adopted by most methods [18, 27, 35]. In the optimization process, each state describes the start-time label of a tube set, and neighboring states are defined as states in which only one active tube changes its start-time
 60 label. The optimization process iterates in a loop of “tube labeling-energy com-

puting” until the energy function reaches its minimum, where the final state of the tube set is determined. It can be seen that any change of a tube’s label leads to re-computation of the energy function, consisting of several cost terms. Obviously, the computation cost is proportional to the number of tubes and the mapping labels. As a result, minimizing such a problem is very time-consuming, especially when the number of tubes or mapping labels is very large.

The goal of video synopsis is to pack all the tubes into a shortened video while avoiding collisions as much as possible. By analyzing the 3D trajectories of tubes in the spatial-temporal domain, it is observed that the underlying reason for tube collisions is that they pass the same area in the image domain at the same time. Therefore, to prevent possible tube collisions when generating synopsis video, one should guarantee that tubes passing the same area of image domain at different times. Moreover, the fact whether or not the tubes will collide in the synopsis can be determined beforehand. We thus propose to rearrange the tubes in a different way. Unlike traditional methods that directly label the start-time of each tube, we position the tubes in an indirect way by first finding the potential collision points between tubes and then labeling the collision points to make sure that collision points of tubes have different time labels in the synopsis. As a result, the process flow for tube rearrangement is changed into “potential collision point locating→potential collision point labeling→tube’s start time labeling” instead of a loop adopted by the traditional methods. Compared with other methods, the main advantage of this new process flow is that it computes the inter-tube collision relationship only once.

The basic idea of our approach is that the tubes and the potential collision relationships among them in a video can be efficiently represented by a graph, and hence, by coloring the graph, the tube rearrangement can be solved easily and effectively. Figure 1 summarizes the overall procedures of our framework. The framework initially extracts all the tubes and then transforms them into a graph. In the graph, the main nodes (m -nodes) indicate tubes extracted from the video, which is denoted by the circle in the graph model. And the sub-node (s -node) on each main node indicates the potential collision point of the

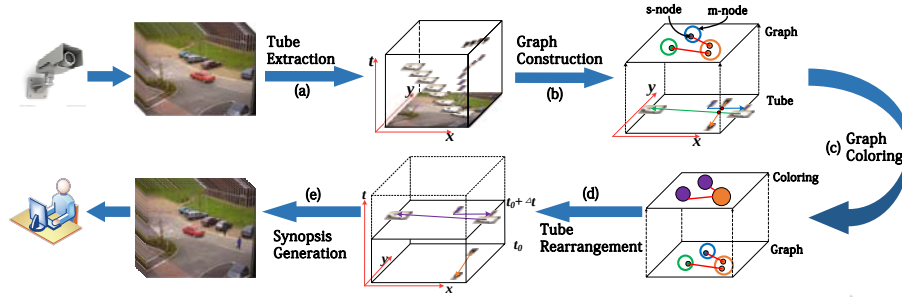


Figure 1: Overview of our framework. (a) Tube extraction from the input video captured by surveillance camera. (b) Graph construction by projecting the tubes into spatial domain and analyzing their potential collision relationship. (c) Graph coloring of *s-nodes*, which represents the potential collision frame in tubes. (d) Tube rearrangement by separating the different colored *s-nodes* of appropriate frames. (e) Synopsis generation for reducing human labor.

tube with another tube, which is denoted by the dot inside the circle in the graph model. The attributes of *m-node* record the tube index and its appearing time, while the attributes of *s-node* record collision times of the tube with other tubes. The edges between *s-nodes* belonging to different *m-nodes* are connected if potential collision relationship exists, and these relations can be defined and determined by analyzing the geometry relationships between tubes in 3D space. To avoid collisions in the synopsis, we have to guarantee connected *s-nodes* have different time labels, which are different colors in graph coloring theory. In this way, the tube rearrangement problem is formulated as a graph coloring problem and can be solved by using state-of-the-art graph coloring algorithms, e.g., [36].

Once the *s-nodes* are colored with different labels, the start-time label of each main-node, that is a tube, can be easily calculated from their node attributes, so that the tube can be accordingly shifted and fused into the background to produce the final video synopsis.

The main contributions of this work are summarized as follows:

- We analyze the underlying reason of collisions happening in the synopsis video and define three kinds of relationship between tubes to describe their potential collisions.

- 110 • We construct a graph structure to represent the tubes and their potential relationships and formulate the tube rearrangement as a graph coloring problem.
- To further mitigate the collision artifacts in the synopsis video, we adopt the more powerful $L(q)$ -coloring [36] of graph where the parameter q is left for user adjustment. Moreover, we design a greedy algorithm to solve 115 the $L(q)$ -coloring problem.

The rest of the paper is organized as follows. Section 2 outlines some related works. Section 3 introduces the details of graph construction and coloring algorithm. Section 4 presents the flow of synopsis video generation. Section 5 120 shows the experimental results. Finally, Section 6 concludes the paper.

2. Related Work

In this section, we review the most related work in video synopsis, particularly the object-based synopsis, and discussion about frame-based abstraction can be found in [6, 7, 8]. Previous work on graph coloring problem and its 125 applications are also outlined.

2.1. Object-based synopsis

The object-based synopsis is able to condense an original video in both temporal and spatial domains. Among all these methods, the proposals in [15, 18, 19, 20] are pioneering works. Kang *et al.* [15] explicitly extracted 130 informative space-time video portions which can be moved and stitched using first-fit and graph-cut optimizations to maximize the amount of visual information. As boundaries between portions exist, their approach would result in obvious seams in the generated video. Rav-Acha *et al.* [20] proposed an object-based approach for video synopsis that is similar to that of Kang *et al.* [15] but 135 only moves objects along the time axis. Pritch *et al.* [19] extended the work in [20] to process always-on videos captured by surveillance cameras or webcams. The more complete work is presented in [18], which includes an online phase

and a response phase. The online phase is responsible for tube extraction and background images recording, whereas the response phase rearranges the tubes
 140 along the time axis to determine the display time of moving objects in a synopsis. It can handle always-on videos and take illumination-varying backgrounds into account. With the similar idea, Xu *et al.* [23] proposed an object-based method of the video synopsis. They considered an object as a spatial-temporal collection, and rearranged objects by maximizing visible information to shorten
 145 the video.

Following the framework in [18], several improved methods in speed or tube rearrangement have been proposed. To better utilize the video space in the spatial domain, Nie *et al.* [27] proposed to globally shift moving objects in both spatial and temporal domains, which shifts objects temporally to reduce the
 150 length of the video and to shift colliding objects spatially to avoid visible collision artifacts. Besides, they used a multi-level patch relocation (MPR) method to construct an expanded background based on environmental content to fit with the shifted objects. To speed up the synopsis generation and achieve real-time performance, online methods have been proposed. In [25, 26], the synopsis
 155 video generation problem is formulated as a maximum a posteriori probability (MAP) estimation problem, where the positions and appearing frames of video objects are chronologically rearranged in real time according to an online updated synopsis table without any optimization procedure. It significantly reduces the computational complexity and can be applied to online streaming videos.
 160 However, since the method does not screen the entire video in advance to obtain the complete trajectories, it only makes sure the initial temporal locations of tubes do not collide. It cannot guarantee avoiding collisions for the subsequent positions of tubes. Another excellent online framework was proposed by Zhu *et al.* [32, 33], which transforms the optimization problem of tube rearrangement
 165 into a stepwise optimization problem and uses Graphic Processing Unit (GPU) and multicore techniques to accelerate the speed. It is able to condense video with less memory and higher speed and is suitable for real-time endless surveillance videos. Other methods try to improve traditional synopsis method from

different aspects, such as adopting novel tracking strategy [28], extracting perfect tubes [29, 31], preserving motion structure [30], and generating abnormal activity video synopsis [21].

2.2. Graph coloring

Graph-coloring is a well-known combinatorial optimization problem which belongs to the class of NP-hard problems [37]. It investigates the ability to efficiently partition the elements of a graph so that no two adjacent or incident elements share the same color, which includes three types coloring of elements: nodes only, edges only, or both nodes and edges. Here, we only consider coloring of the nodes.

There are many real-world applications that can be modeled by graph coloring, such as planning and scheduling [37], pilot allocation in massive MIMO systems [33], timetabling [38, 39], channel assignment in cellular network [40, 41, 42, 43], radio frequency assignment [44], protein-protein interaction (PPI) networks [45], and so on. In the following typical applications are outlined.

To solve the Bin Packing Problem with Conflicts (BPPC), several methods transform it to graph coloring problem [46, 47]. Gendreau [46] firstly used the graph theory to represent the conflict relationship, and designed a gradual approximate algorithm through computing maximum induced sub-graph iteratively. Subsequently Ye, *et.al* [47] transformed the model of Bin packing problem with conflicts (BPPC) into the representation of conflict graph structure and colored it using an ant colony optimization (ACO) algorithm to eliminate conflicts between items. Experiments validate the feasibility and efficiency of this method for the BPPC.

In communication areas, Hale [44] formulated the radio channel assignment problems in wireless communication as the graph T-coloring problems. The model with graph theory makes each transmitter correspond to a vertex, with edges corresponding pairs of nearby transmitters. The original T-coloring is that the absolute difference between channels for two nearby transmitters must not belong to a certain set T and to minimize the span of the assignment, as

well as to minimize the order of the assignment, which is the number of the integer(frequency) channels that the assignment uses. In [33], a graph coloring based pilot allocation (GC-PA) scheme was proposed to mitigate pilot contamination for multi-cell massive MIMO systems. They firstly construct an interference graph to describe the potential ICI relationship of all users and then allocate the limited pilot resource by solving the coloring problem of the graph. In [48], the authors formulated the spectrum sharing problem in cognitive MIMO systems as a new vertex coloring problem and designed an efficient greedy algorithm to find the optimal coloring solution. It turns out that their proposed algorithm can approach the maximum achievable sum rate. To further prevent interferences between channels, Griggs and Yeh [36] introduced integer $L(2, 1)$ -coloring of a graph G , and proposed $L(k_1, k_2, \dots, k_p)$ -coloring problem of a graph, where the labels of neighboring vertices differ by at least k_1 and the labels of vertices at distance i differ by at least $k_i (i = 1, 2, \dots, p)$. Such a generalization of graph coloring makes it contain more constraints. Many authors have subsequently contributed to $L(2, 1)$ -colorings, and recently $L(k_1, k_2, \dots, k_p)$ -colorings. Survey papers on applications of graph coloring can be found in [49].

Concerning the solutions of graph coloring problem, many optimization algorithms can be used, such as the Saturation algorithm, Recursive Largest First algorithm, Degree of Saturation Algorithm, Simulated Annealing algorithm, Greedy algorithm, and many others [50]. Recently, parallel graph coloring algorithms [51, 52] and implementation on the GPU platform [53, 54] have also been proposed to improve the real-time performance of graph coloring. However, the graph coloring algorithms are not the main topic of this paper and interesting readers can refer to [55, 56, 57, 58] for thorough discussion.

To the best of our knowledge, graph coloring has no applications in the field of video synopsis so far, although graphs have been frequently utilized to represent events in videos [27, 59], and graph cuts method is adopted to do the energy minimization procedure in [27]. In this paper, we explore the application of graph coloring in video synopsis and solve the tube rearrangement, the critical stage in video synopsis generation, by formulating it as a graph coloring problem.

230 3. Graph Construction and Coloring

In this section, we first analyze the potential collision relationship between tubes extracted from the video. Then the construction process of potential collision graph from the tubes is elaborated based on the analysis. Finally, a greedy algorithm for coloring the graph to rearrange the tubes appropriately is presented and discussed in detail.

3.1. Relationship between Tubes

The original video with N frames is represented in a 3D space-time volume as $I(x, y, t)$, where (x, y) is the spatial coordinates of the pixel and t ($1 \leq t \leq N$) is the frame number. A tube is a frame sequence of the same moving object and can be depicted in a 3D spatial-temporal volume. The base of the volume cube represents the x - y spatial dimension (image domain) and the cube's height represents the temporal dimension (time domain), as shown in Figure 2. The extraction of tubes will be described in detail in Section 4.1.

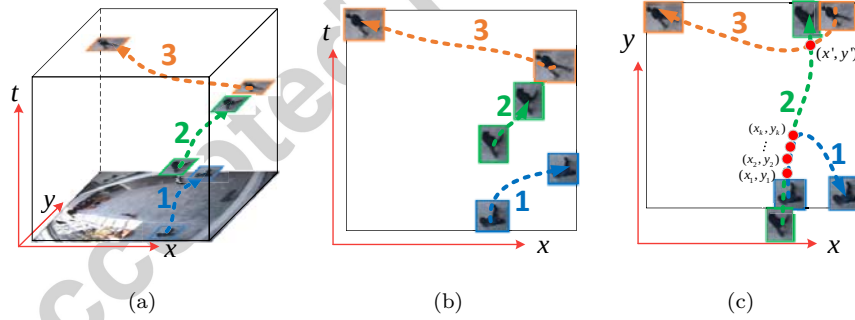


Figure 2: The video volume with three tubes. (a) Three tubes in 3D spatial-temporal domain. (b) The 2D x - t axis mapped from (a) to present the tubes in time-axis. (c) The 2D spatial domain of (a), where the red dots represent the potential collision parts of tubes when rearranging the tubes in synopsis.

Let T_i be the tube of a dynamic object with index i and its related information is obtained from a tracking procedure. During the tube's existing interval $t_i^s \leq t \leq t_i^e$, a sequence of rectangle bounding boxes R_i^t are used to denote

its spatial locations formally. Although the term ‘tube’ and ‘trajectory’ are used interchangeably in some similar works, we differentiate the two terms here. The trajectory of an object is depicted by the central points of a sequence of rectangle bounding boxes R_i^t while the tubes are represented by the rectangle bounding boxes $T_i = ([t_i^s, t_i^e], \{R_i^t\})$, where t_i^s and t_i^e stand for the starting-time and the ending-time of tube i , respectively.

To generate video synopsis, the existing methods usually shift the tubes in time dimension only [19] or in both time and spatial dimension simultaneously [28]. However, collisions may occur in the generated synopsis when the tubes are shifted inappropriately. It is easy to see that from Figure 2(a), the underlying reason for unpleasant collisions in synopsis is that two tubes intersect or overlap in the 3D space, *i.e.*, two objects are rearranged to pass the same area in the image domain at the same time. Therefore, to mitigate or eliminate collisions, one should guarantee that no two tubes intersect or overlap in the 3D space-time volume. However, traditional methods usually judge two tubes collide or not and how seriously they collide in the 3D space iteratively during the whole optimization process: the judgment is conducted repeatedly once a tube changes its synopsis label, which causes large amounts of computation redundancy. Essentially, the fact whether or not two tubes collide in the final synopsis can be determined beforehand by projecting them onto the image plane. As shown in Figure 2(c), tube (green) and tube (yellow) intersect at point (x', y') , while tube (green) and tube (blue) overlap at a series of points $(x_1, y_1), \dots, (x_k, y_k)$. These points indicate potential collisions of the tubes in synopsis and will become real ones if they are rearranged to pass these points at the same frame, whereas other parts of tubes will never cause collision at all. Accordingly, much emphasis shall be put on these potential collision points instead of the whole tubes. To avoid collisions happening in the synopsis, we should guarantee that tubes are passing these potential collision points at different times. Below three types of relationship between tubes will be defined to judge whether two tubes collide or not before rearranging them.

Given two tubes i and j , three types of relationships between them can be

defined according to their spatial relationship after projecting them onto the image plane, as shown in Figure 3.

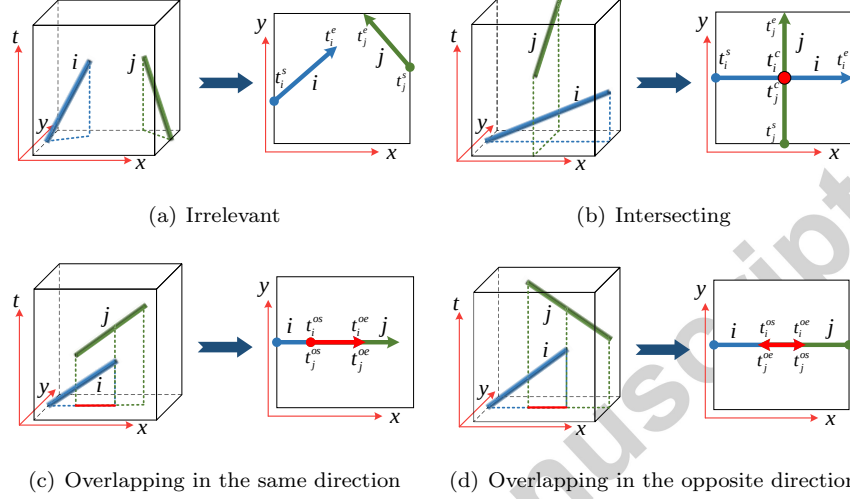


Figure 3: Relationship between tubes. For the sake of visual quality, we draw the tubes as straight lines. However, this does not affect the determination of relationship between non-straight tubes as discussed in the paper.

a) **Irrelevant.** Two tubes are said to be irrelevant if they have no intersecting points after being projected onto the image plane, *i.e.*, the x - y plane as shown in Figure 3(a), where t_i^s , t_i^e and t_j^s , t_j^e are the starting and ending frame of tube i and tube j , respectively. In this case, the two tubes don't pass the same area of the image plane at all during their appearance in the camera view, which means that they will never cause collision when being shifted along the time axis in synopsis video. As a result, tubes with this kind of relationship can be rearranged free of collisions arbitrarily, if other issues, such as chronological order and so on, are not considered.

b) **Intersecting.** Two tubes are said to be intersecting if their projections onto the image plane intersect, as shown in Figure 3(b). Although the two tubes don't meet in the original video volume, as shown by the left figure of Figure 3(b), they probably cause collision if they are rearranged improperly. Specifically, let t_i^e and t_j^e be the frame number of the intersection point of tube

i and j , respectively, collision will definitely occur if t_i^c and t_j^c are labeled with
 295 the same frame-label in the synopsis video.

c) **Overlapping.** Two tubes are said to be overlapping if their parts or whole tubes share the same spatial regions after being projected onto the image plane, as shown in Figure 3(c) and Figure 3(d). This corresponds to the case that the original tubes are in the same plane of the 3D spatial-temporal volume which
 300 is perpendicular to the image plane. Two different kinds of relationship exist in this case: overlapping in the same direction and in the opposite direction, as shown in Figure 3(c) and Figure 3(d), respectively. The overlapping starting frames of tube i and j are denoted by t_i^{os} and t_j^{os} , and the overlapping ending frames of them are represented by t_i^{oe} and t_j^{oe} . Note that t_i^{os} , t_i^{oe} , t_j^{os} and
 305 t_j^{oe} are all ordinal numbers in frame counting from the tubes' starting time. This is the most difficult case for tube rearrangement and is prone to cause serious collisions in the synopsis video, especially for the case of overlapping in the opposite direction. Moreover, it is also one of the factors that affects the condensation ratio of synopsis since there is not too much free space for the
 310 objects to shift in time dimension without causing unpleasant collision artifacts. To obtain more condensed synopsis video in this case, a possible solution is to shift one of the two tubes in a parallel fashion to make sure they don't overlap on the image plane, *i.e.*, they don't lie in the same plane perpendicular to the image plane the 3D space-time volume. However, we think that shifting the
 315 tubes in spatial domain may damage the authenticity of original scenes and thus still choose to shift tubes along time axis only as most works did.

Having relationships between tubes defined, the remaining problem is how to determine their relationship and how to compute their intersecting point or overlapping parts. Here, the relationship between tube i and tube j are
 320 determined by computing the overlapping areas of their bounding boxes $\{R_i^t\}$ and $\{R_j^t\}$ on the image plane and counting the frames. However, in real scenes, two tubes may not exactly intersect at just one frame on the image plane for intersecting case; they also have several frames overlapping. Therefore, it is necessary to differentiate the relationship of intersecting and overlapping. Here,

we adopt an overlapping threshold to differentiate the two types of relationships by counting two tubes' overlapping frames F_i and F_j : it is defined as overlapping if either F_i and F_j is greater than the overlapping threshold; otherwise it is defined as intersecting. For tubes defined as intersecting, the central frame of their overlapping frames is deemed as the intersecting point. The overlapping threshold is set as 5 frames in our paper, which is obtained in a trial and error way. Moreover, related information of tubes potential collision points, such as and for intersecting tubes (Figure 3(b)), and t_i^{os} , t_i^{oe} , t_j^{os} , t_j^{oe} for overlapping tubes(Figure 3(b)) should be recorded for subsequent use.

3.2. Potential Collision Graph Construction

Having determined the relationships between tubes, we now introduce the construction procedures of potential collision graph, which is the pivotal data structure of the framework in this paper. It not only stores the information of tubes, but also records the relationships between them. The formal definition of potential collision graph is described as follows.

Definition 1: Potential collision graph (PCG) is a hybrid graph structure $G = (V, E)$, which represents the potential collision relationship among the tubes. More specifically, the nodes set V corresponds to the tube set and the edges set E corresponds to the potential collision relationship among the tubes.

The construction of *potential collision graph* mainly consists of two steps: node abstracting and edge weighting.

3.2.1. Abstracting the Nodes

Unlike the traditional methods that usually designate nodes to represent events or tubes in the video, the PCG consists of two kinds of nodes: main-nodes (m -nodes) and sub-nodes (s -nodes). An m -node represents a tube extracted from the original video, and an s -node on the m -node stands for one potential collision point on the tube interacting with another tube. In this sense, the m -node and s -node form a parent-child relationship.

Figure 4 describes the entire process of **node abstracting**, which primarily consists of three steps: tube extraction, **relationship analysis**, and **node generation**. After extracting the tubes from the video, their relationships are analyzed and potential collision points are determined, as shown in Figure 4(b). Then the tubes are represented by *m-node* and collision points are denoted by *s-nodes* respectively, as shown in Figure 4(c), v_B^m is an *m-node* representing tube B and v_B^s is an *s-node* on v_B^m abstracted from the potential collision point (See Figure 4(b)). Specifically, for intersecting tubes, two *s-nodes* are abstracted from one collision point, one *s-node* for each tube, see v_B^s and v_C^s in Figure 4(c) for example. For overlapping tubes, four *s-nodes* for each tube are abstracted, two *s-nodes* for each tube, indicating the starting and ending of overlap happening, see v_{Ds}^s, v_{De}^s on v_D^m and v_{Es}^s, v_{Ee}^s on v_E^m . Moreover, the related information of tubes and collision points are recorded in the attributes of nodes. For example, the information about the tube, such as starting-time, ending-time, bounding boxes, *i.e.*, $T_A = ([t_A^s, t_A^e], \{R_t^A\})$ is taken as the attributes of corresponding *m-node*. Accordingly, the attribute of *s-nodes* records related information of corresponding collision points. For intersecting tubes, the frame number of intersecting point is recorded as attributes of the abstracted *s-nodes*, for example, t_B^c for *s-node* v_B^s and t_C^c for *s-node* v_C^s . For overlapping tubes, such as tube D and tube E in Figure 4(b), two points' frame numbers need to be recorded for each tube: the starting and ending frame of overlap happening. As shown in Figure 4(c), t_D^{os} and t_D^{oe} are recorded for *s-node* v_D^s , and t_E^{os} and t_E^{oe} are recorded for *s-node* v_E^s respectively.

3.2.2. Weighting the Edges

To finish the potential collision graph construction, edges should be connected between the nodes. Three kinds of edges are designed to fulfill our goal of tube rearrangement in synopsis generation: edges between *s-nodes* on an *m-node*, edges between *s-nodes* of different *m-nodes* with intersecting relationship, and edges between *s-nodes* of different *m-nodes* with overlapping relationship.

In order to differentiate these three kinds of edges, different weights are

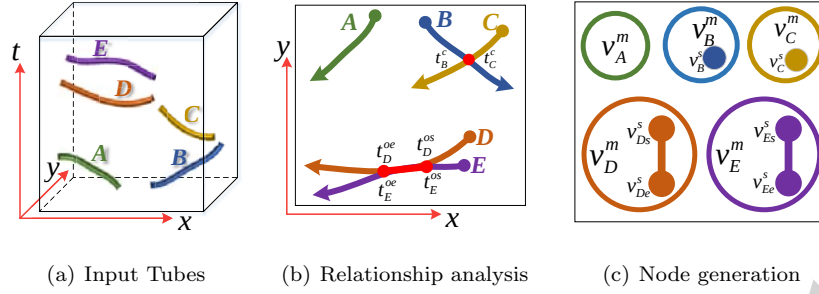


Figure 4: Abstracting the nodes.

assigned for the edges.

a) Directed edges between *s-nodes* on the same *m-node* are weighted using their interval in original video with direction pointing from starting-frame to ending-frame, see $\Delta t_D = t_D^{oe} - t_D^{os}$ and $\Delta t_E = t_E^{oe} - t_E^{os}$ in Figure 4(b) for example, which ensures the whole tube moves in timeline as a whole without causing contradicting labeling (see next section for detail).

b) Undirected edges between *s-nodes* of different *m-nodes* with intersecting relationship are weighted by 1, indicating potential collisions exists between the two tubes at the *s-nodes* and attention shall be paid on these *s-nodes* to prevent potential collisions from becoming real ones. For example, we weight the edge between the *s-nodes* t_B^c and t_C^c as $e_B^C = 1$.

c) Undirected edges between *s-nodes* of different *m-nodes* with overlapping relationship are weighted by using negative numbers to differentiate the relationship from other two kinds of weight. Since four *s-nodes*, i.e. two corresponding pairs, are abstracted in this case, two edges shall be connected for every two tubes with overlapping relationship. For the sake of easy differentiation, we add the same weight for the two edges corresponding to the same overlapping tubes, and give different weights for different overlapping tubes. For example, the edges between the *s-nodes* v_D^{os} and v_E^{os} can be weighted as $e_D^E = -1$, the same weight should also be added to the edge between the *s-nodes* v_D^{os} and v_E^{os} to indicate that they correspond to the same overlapping tubes. For other overlapping tubes, different negative numbers will be used to weight the edges.

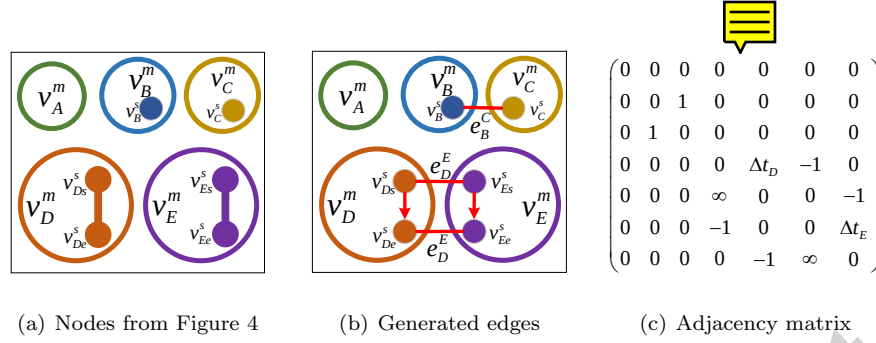


Figure 5: Process of weighting the edges.

405 Tubes having irrelevant relationship with other tubes will be left unconnected, and become isolated m -nodes. Moreover, an adjacency matrix, $A = (a_{ij})_{m \times m}$ where m is the total number of s -nodes and isolated m -nodes, is used to store the PCG structure so that the information in the PCG can be better represented and easy utilized. Figure 5 illustrates the process of weighting the edges.

410

3.3. Graph Coloring for Tube Rearrangement

To prevent potential collisions between tubes from becoming real ones when rearranging the tubes in the final synopsis video, different tubes should go through the potential collision points at different times that are sufficiently far apart. Since we model the potential collision points as s -nodes in the PCG and use edges with different weights to connect them, tubes go through these collision points at different times means that the s -nodes with connecting edges should be colored with different colors, i.e., different time labels in the synopsis video. Besides, video synopsis aims to pack tubes as more as possible in as shorter interval as possible, whereas graph coloring is to color the graph using the minimum possible number of colors. Therefore, the tube rearrangement in synopsis generation can be formulated as a typical graph coloring problem, to be more specifically, node coloring problem. However, only one color (i.e., frame) separation is not a safe distance in timeline to avoid collisions by considering the sizes and speeds of moving objects. We thus use the generalization of graph

420

425

node coloring problem where more restrictions can be placed, the $L(q)$ -coloring, which is defined as follows.

Definition 2: $L(q)$ -coloring. Given a graph $G = (V, E)$ and a nonnegative integer q , an $L(q)$ -coloring is an assignment of nonnegative integers to the nodes of G . Adjacent nodes are labeled by using colors at least q apart and the total colors should be as less as possible. That is to find a coloring function $f_q : V \rightarrow N$ such that $|f(u) - f(v)| \geq q$ if $u, v \in E$, while the difference between the largest and the smallest used colors is minimal. The minimum span over all possible labeling functions is denoted by $\lambda_q(G)$ and is called λ_q number of G .

Obviously, the classical node coloring is a special case of $L(q)$ -coloring when the parameter $q = 1$. Since the PCG is a hybrid graph structure consisting of m -nodes and s -nodes, more constraints need to be imposed on the traditional $L(q)$ -coloring to meet our requirement on tube rearrangement. Specifically, the constraints are as follows.

1) Only s -nodes on connected m -nodes (see v_B^s on v_B^m and v_C^s on v_C^m in Figure 5) and isolated m -nodes (see v_A^m in Figure 5) are colored, whereas connected m -nodes are not colored. The connected m -node is used to ensure its s -nodes not being contradictorily colored. For example, let $l(\bullet)$ denote the assigned color for nodes, the colors for v_{Ds}^s and v_{De}^s on v_D^m (see Figure 5) should satisfy $l(v_{De}^s) - l(v_{Ds}^s) = t_D^{ce} - t_D^{cs}$. In other words, once one of the s -nodes on the same m -node is colored, colors for other s -nodes on the same m -node can be deduced from their time distance in original video. (see v_{Ds}^s and v_{De}^s on v_D^m and v_{Es}^s and v_{Ee}^s on v_E^m in Figure 5). This is to make sure that the tubes or parts of the tubes are not being lengthen or shorten during the rearrangement. Otherwise, it may produce unnatural artifacts.

2) Besides that the adjacent nodes shall be colored by q colors apart, extra constraints should be imposed on the four nodes connected with edges of the same negative weights, i.e. tubes with overlapping relationship. See Figure 3(d) for example, if we color the s -nodes $v_{is}^s, v_{ie}^s, v_{js}^s, v_{je}^s$ abstracted from tube i and j with $l(v_{is}^s), l(v_{ie}^s), l(v_{js}^s), l(v_{je}^s)$, the constraint to ensure no collision in this

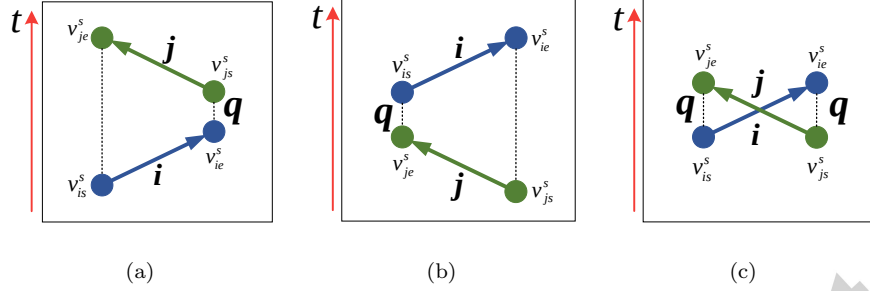


Figure 6: Coloring constraints for overlapping tubes. Tube rearrangement satisfying the constraints in Eq.(1) is shown in (a) and (b), which guarantees no collision happening, while the collision situation violating the constraints is shown in (c).

case is

$$[l(v_{is}^s) - l(v_{je}^s)] \bullet [l(v_{ie}^s) - l(v_{js}^s)] > 0 \quad (1)$$

as shown in Figure 6, otherwise intersections happen between tubes which definitely cause collisions.

Since graph node coloring is NP hard, there is a need of heuristic methods to determine the suboptimal solutions in polynomial time. Here we are proposing heuristic algorithm for $L(q)$ -coloring in our framework that can run fast and provide a good quality solution. It follows the idea of the Degree of Saturation (DSATUR) algorithm [60] and Recursive Largest Fit (RFL) algorithm [61]. In DSATUR algorithm, the nodes are ordered by a decreasing order of saturation, where the saturation degree is determined as a number of different colors to which a node is adjacent. In Recursive RFL algorithm, each color is completed before introducing a new one so that nodes of one class obtain colored at a time. In our algorithm, we combine both the ideas of DSATUR and RFL. The algorithm iterates to color the nodes as the value of color increases from the minimum value until there are no nodes left uncolored. In each iteration, the nodes are sorted in the decreasing order of saturation degree and the node with the largest saturation degree is firstly colored with the current color. To minimize the number of colors used, one color in each iteration should be used to color as many nodes as possible.

In DSATUR algorithm, the saturation degree is a term for evaluating the difficulty of coloring certain node and the most difficult one is colored with priority. In our situation, the colors of *nodes* actually stand for the positions in time dimension where the corresponding tubes will be placed. Therefore, characteristics about the tubes shall be taken into account in computing the saturation degree, such as the length of the tube, the appearing time of the tube, and the number of potential collisions of the tube with other tubes. Obviously, the longer the tube and the earlier the tube appears, the smaller time-label it should be given in the synopsis video for the sake of synopsis length and chronological order. Besides, the more the potential collisions the tube intersects with other tubes, the more difficult it is to assign a time-label for the tube without causing collisions. That is to say, the nodes standing for the earlier appearing, longer tubes with more potential collisions will be colored firstly since the color increases from the minimum value. Therefore, the saturation degree of the *s-node* v_i^s is calculated as:

$$S_{v_i^s} = S_d(v_i^s) + S_l(v_i^s) + S_{app}(v_i^s) + S_{pc}(v_i^s) \quad (2)$$

where $S_d(v_i^s)$ is the traditional saturation degree, $S_l(v_i^s)$, $S_{app}(v_i^s)$, and $S_{pc}(v_i^s)$ are new items added for our application accounting for the length, appearing time, and number of potential collisions of the tube, respectively.

Assume the *m-node* that *s-nodes* v_i^s belongs to is denoted as v_i^m and its corresponding tube is $T_i = (t_i^s, t_i^e, \{R_t^i\})$, then the tube's length is $L_i = t_i^e - t_i^s$. The term $S_l(v_i^s)$ and $S_{app}(v_i^s)$ are computed as:

$$S_l(v_i^s) = \frac{L_i}{\max_{k=1, \dots, M} \{L_k\}} \quad (3)$$

$$S_{app}(v_i^s) = \frac{N - t_i^s}{N} \quad (4)$$

where N is the number of frames of the original video and M is the number of tubes extracted from the video. For computing $S_{pc}(v_i^s)$, we count the number of *s-nodes* that are on the same *m-node* with v_i^s and check the weight of edge

that v_i^s is connected by:

$$S_{pc}(v_i^s) = Num(v_i^m) + \sum_{a_{ik} \neq 0} w_k$$

$$w_k = \begin{cases} 1, & \text{if } \text{sgn}(a_{ik}) = 1 \\ 2, & \text{if } \text{sgn}(a_{ik}) = -1 \end{cases} \quad (5)$$

where $Num(\bullet)$ stands for the number of s -nodes that are on the same m -node, and w_k is the weight. Here, we give larger weights to the s -nodes with negative weighted edges than those with positive weighted edges to give high priority to tubes with overlapping relationships, because tubes with overlapping relationships is prone to cause serious collisions and difficult to deal with. For isolated m -node v_i^m , we set the terms $S_d(v_i^m)$ and $S_{pc}(v_i^m)$ to 1 to give isolated m -nodes higher priority because isolated m -nodes never cause collisions with others and can get any color. Here, we choose to give them colors with the minimum value.

As can be seen from the Eqs.3 and 4, the terms $S_l(v_i^s)$ and $S_{app}(v_i^s)$ are already normalized into the range $[0, 1]$, while the terms $S_d(v_i^s)$ and $S_{pc}(v_i^s)$ are not. For fairness, these two terms are normalized by the maximum value of all the nodes in the PCG before they are added together.

With the definitions above, we now introduce our $L(q)$ -coloring algorithm of the PCG. Let $v_1, v_2, \dots, v_n$ be all the s -nodes and isolated m -nodes in the PCG, given the integer color labels $l = 1, 2, 3 \dots$, the objective of $L(q)$ -coloring of the PCG is to color the nodes with as less colors under the following constraints: (1) nodes connected by undirected edges shall be colored by q apart; (2) nodes connected by negative undirected edges shall further satisfy Eq.1. The proposed $L(q)$ -coloring algorithm of the PCG consists of the following steps:

1) Initialization:

- Initialize the color to the minimum value.
- Define a type flag f_i , $i = 1, 2, \dots, n$ for each node to indicate its type:
 m -node or s -node, $f_i = \begin{cases} 1, & v_i \text{ is } s\text{-node} \\ -1, & v_i \text{ is isolated } m\text{-node} \end{cases} \quad i = 1, 2, \dots, n.$

- Define a flag f_i^c , $i = 1, 2, \dots, n$ for each node to be colored and initialize $f_i^c = 0$, $i = 1, 2, \dots, n$.
- Compute the saturation degree using Eq.2;

2) Iterative coloring:

- 530
- Dynamically order the nodes with decreasing saturation degree
 - Color the nodes in the decreasing order of saturation degree until no nodes can be colored with the current color: if it is an *s*-node, update the colors of other *s*-nodes that are on the same *m*-node with it according to the directed edges in $A = (a_{ij})_{n \times n}$.
- 535
- Increase the color by 1 and repeat the above three sub-steps.

The pseudo code is summarized in Algorithm 1.

In line 4 of Algorithm 1, although the uncolored nodes need to be dynamically sorted according to their saturation degree, only the term $S_d(v_i)$ in the saturation degree S_{v_i} need to be recomputed and updated, whereas other terms

540 are computed from the characteristics of the *m*-node that the *s*-node belongs to. There is no need to recompute them during the iteration. In the case of ties in saturation degree between *s*-nodes, the *s*-node with earlier appearing time is ordered in the former place.

In line 5 of Algorithm 1, the **Cond1** is to make sure that the adjacent nodes

545 are colored with colors at least q apart, and **Cond2** is to impose the constraints in Eq.1. They are defined as follows:

Cond1: For node v_i , all the nodes connecting with v_i by edge weighting 1 are not using colors in $[l - q, l + q]$;

Cond2: For node v_i , if there is an *s*-node v'_i connecting to it by edge weighting -1, find the other corresponding nodes connected by edge weighting -1, say

550 as v_j and v'_j . v_i and v_j belong to the same *m*-node, v'_i and v'_j belong to the same *m*-node. If v'_i and v'_j are already colored, judge $(l - l_{i'}) (l + t_{ij} - l_{j'}) > 0$ is true or not; if v'_i and v'_j are not colored yet, this condition is always true. $l_{i'}$, $l_{j'}$ are the colors for nodes v'_i and v'_j , t_{ij} is the distance in frame between v_i and v_j .

Algorithm 1 $L(q)$ -Coloring of the PCG for tube rearrangement

Input: The s -nodes and isolated m -nodes in the PCG $v_1, v_2, \dots, v_n$ and the

type flag f_i , $i = 1, 2, \dots, n$;

Adjacency matrix $A = (a_{ij})_{n \times n}$ and parameter q

Output: Colored nodes

```

1:  $l \leftarrow 1, f_i^c = 0, \quad i = 1, 2, \dots, n$ ;
2: while there is any  $f_i^c = 0$  do
3:   Sort the uncolored nodes in the decreasing order of  $S_{v_i}$ ;
4:   for each uncolored node  $v_i$  in the decreasing order of  $S_{v_i}$  do
5:     if Cond1 and Cond2 are all satisfied then
6:       assign color  $l$  to  $v_i$  and set  $f_i^c = 1$ ;
7:       if  $f_i = 1$  then
8:         color other  $s$ -nodes  $v_k$  on the same  $m$ -node with  $v_i$  according
           to the directed weight  $a_{ik}$  in  $(a_{ik})_{n \times n}$  and set  $f_k^c = 1, k = 1, 2, \dots, n$ ;
9:       end if
10:    end if
11:  end for
12:   $l \leftarrow l + 1$ 
13: end while

```

555 Lines 7-8 of Algorithm 1 are to make sure that the s -nodes on the same m -node are not contradictorily colored as talked before.

After coloring the PCG, time labels are given for the s -nodes and isolated m -nodes, *i.e.* the time labels for collision points on connected m -nodes. However, we usually label the start-time in synopsis for each tube and this can be done easily through a simple subtraction step. For example, if we color the s -node (collision point) of m -node (tube B) in Figure 4 by the label l_B^c , the start-time of tube B can be simply computed by the information in the attributes of v_B^s , that is $l_B^s = l_B^c - (t_B^c - t_B^s)$. As the color begins from 1, negative time labels for starting time of tubes may appear after subtraction, which can be solved by a translation along time axis for all tubes.

560

565

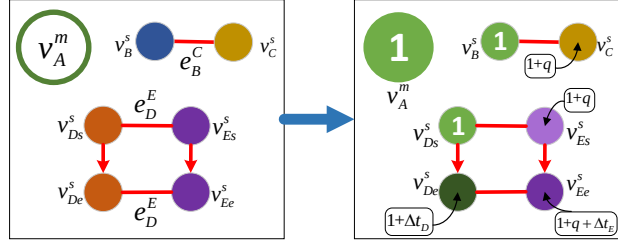


Figure 7: Graph coloring of nodes in Figure 5(b)

The illustrative coloring of graph in Figure 5(b) is shown in Figure 7. In the first iteration, the isolated m -node v_A^m and s -nodes v_B^s , v_{Ds}^s are colored by 1 because they are not connected and have higher saturation degree. Once v_{Ds}^s is colored, the s -node v_{De}^s shall be updated by $1 + \Delta t_D$ accordingly. Then the color increases as the iteration goes on until all the s -nodes are colored without violating the coloring constraints.

3.4. Selection of Parameter q

The parameter q is a key factor which affects the performance of our method by compromising between condensation ratio and collision artifacts. Appropriate setting of this parameter should take into account both the sizes of the objects and the speeds at which they are moving, ensuring their proper separation in timeline between potential collision points. As we mainly concentrate on the site-seeing surveillance video in this paper, this parameter can be set by recording the tubes appearing in the view and analyzing their speeds and sizes statistically. Specifically, we count the frames of different objects passing by a certain pixel in the scenes under different scenarios. Tubes of two kinds of objects, cars and pedestrians obtained by the site-seeing surveillance videos, are extracted to do the statistical analysis. For each kind of objects, more than 5000 tubes are counted for reliability. Figure 8 shows the statistical results for cars and pedestrians, where the horizontal axis stands for the frame number that the objects need to pass by certain pixel completely and the vertical axis records the corresponding number of tubes. It can be seen from Figure 8 that the peak

value lies at 3 for cars and 5 for pedestrians respectively. That is to say, it takes at least 3 frames and 5 frames for most pedestrians and cars respectively to pass by a pixel. Therefore, one frame separation for two tubes between the potential collision points may not guarantee collision free and a larger margin is preferable. This is why we use $L(q)$ -coloring instead of classical graph node coloring in tube rearrangement. The statistical results on cars and pedestrians provide a reference for setting of the parameter q : we can set $q \geq 3$ for cars and $q \geq 5$ for pedestrians. However, different setting of q according to objects' size and speed would add extra computation and may not bring more benefits (see Section 5.1). Therefore, we will set q to a fixed value according to the statistical results in our experiments. In all, the principle for setting of q is, the larger the size and the slower the speed of the object is, the larger the parameter q should be set accordingly.

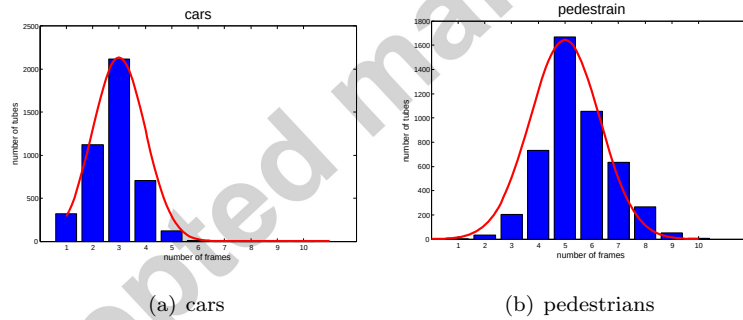


Figure 8: The statistical results for cars and pedestrians.

4. Synopsis Generation

In this section, the steps of synopsis generation are introduced briefly, as shown in Figure 1. Firstly, the input video is analyzed to extract the background and tubes, *i.e.* frame sequences of moving objects. Then, our tube rearrangement and the method to seamlessly fuse the moving objects into the background are discussed.

4.1. Tube Extraction

Since a tube is a frame sequence of the same moving object, its extraction includes two tasks, moving object segmentation and object tracking. With the input video, we first extract background images from it. For most site-seeing surveillance videos, the background is usually static in daily periods with exceptions when illumination varies. Based on this observation, we use the Visual Background Extractor (ViBe) [62] to model the background. The ViBe directly compares the pixel value to the closest values in a sphere centered on the classified pixel value. The radius of the sphere and the threshold of matching number are key to the model accuracy. And the background generation is simply adopted by averaging frames in a fixed time interval. Since the ViBe also affords the use of a strictly conservative update scheme, it exhibits nice robustness to noise.

Next, we explicitly detect and extract the interesting objects, which are tracked by using the linear neighbor object prediction algorithm [63]. According to the fact that the same objects differ with shortest distance between the last and current frame, we construct a matching matrix to represent the relevance of the same object. Define the predicted position of object O_i^t as P_i^t , which represents the bounding box center of object i in frame t and takes direction vector into account. P_i^t is calculated as:

$$P_i^t = P_i^{t-1} + \sum_{n=t_{is}}^{t-1} \lambda^{n-t-1} (P_i^n - P_i^{n-1}) \quad (6)$$

where λ is a parameter less than 1, which satisfies $\sum_{n=t_{is}}^{t-1} \lambda^{n-t-1} = 1$.

Assuming that m object instances in frame t are detected and their corresponding position vectors are computed as $\{N_1^t, N_2^t, \dots, N_m^t\}$, we determine the correct position T_i^t of the tracked object O_i^t by comparing $\{N_1^t, N_2^t, \dots, N_m^t\}$ with P_i^t and find the one with the shortest distance:

$$T_i^t = \{N_a^t | P_i^t - N_a^t = \min_{a,n \in [1,m]} (P_i^t - N_n^t)\} \quad (7)$$

Linear prediction based object tracking algorithm is simple to implement and provides good real-time performance in practice. However, it only can deal with

object tracking under simple scenes and may lead to event subsection problem in
 635 some scenarios. To deal with this problem, we compare the appearance relevance
 among objects to connect fractured trajectory to optimize the tracking result.

In most cases, not all objects are of interest. For example, a traffic surveil-
 lance camera may be interested only in cars, while other applications may prefer
 pedestrians. Therefore, once the tubes are extracted from the input video, we
 640 can select all the tubes or just a fraction of the tubes according to customized
 requirements.

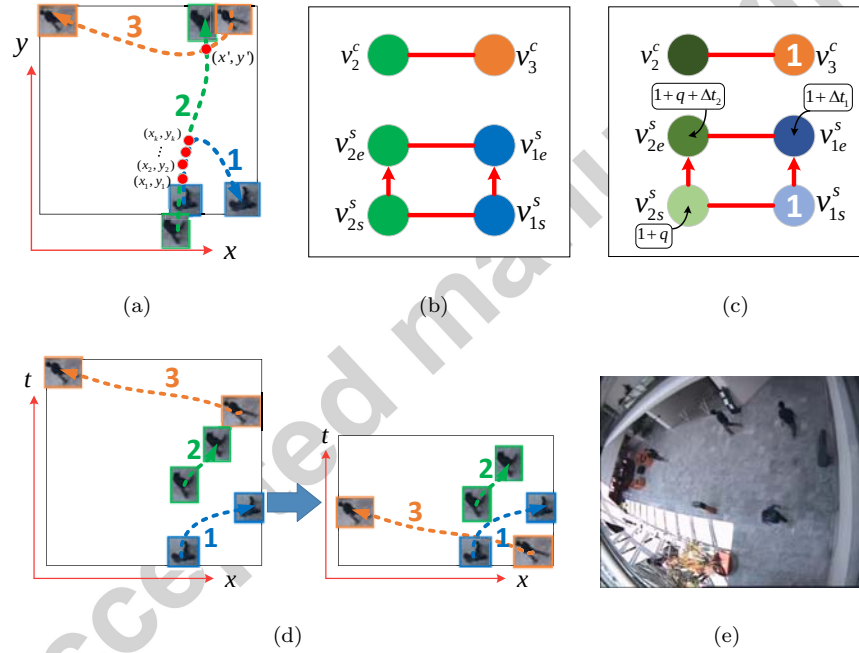


Figure 9: Synopsis generation of the scene in Figure 2. (a) Extracted tubes and their potential collision relationship. (b) Constructed graph. (c) Graph coloring. (d) Illustration of condensation in time domain. (e) One frame of the generated synopsis.

4.2. Synopsis Generation

When the tube sequences are extracted, the information about each tube,
 such as the starting time, ending time, and bounding boxes are recorded. Then
 645 the tubes are projected onto the image plane for relationship analysis, based

on which the potential collision graph is constructed. $L(q)$ -coloring is implemented on the graph to achieve the positioning of tubes in the final synopsis, as described in Section 3. Finally, we use the Poisson image editing [64] for video synopsis generation, which performs well for stitching the rearranged tubes into background images. Figure 9 shows an illustrative example of synopsis generated by our method, which is implemented on the scene shown in Figure 2. Three tubes are extracted and represented by the PCG with m -nodes and s -nodes, as shown in Figure 9(a) and 9(b). Then, coloring of the graph is implemented for tube rearrangement, as shown in Figure 9(c). According to the labels, the tubes are rearranged in the synopsis according to the labels determined by the coloring procedure. Objects appearing at different times in the original video may appear simultaneously in the same frame in synopsis. As a result, the length of the input video will be condensed, as shown in Figure 9(d). Finally, one frame of the generated synopsis is shown in Figure 9(e).

5. Experiments

In order to evaluate the performance of the proposed method, nine surveillance video sequences taken from indoor and outdoor scenes with different resolutions are used, including scenarios of 1 cross road, 3 sidewalks, 1 park, 2 streets, and 2 indoors. In these scenarios, the moving objects contain pedestrians, motorbikes, cars, and so on, among which different relationships such as irrelevant, intersecting, and overlapping exist. The characteristics of the video sequences are listed in Table 1, and the details of the running environment are listed in Table 2.

In the following, the performance of the proposed approach is compared in four aspects: 1) visual quality, 2) frame condensation ratio, 3) compact ratio, and 4) overlap ratio. The visual quality is a subjective metric for user visual pleasure while the other three metrics are subjective metrics for objective comparison with other methods.

- 1) *Frame condensation ratio (FR)*. The FR is defined as the ratio of the

Input Video	Resolution	T_I	N_o	Relationships among Objects	T_S
T1 Cross Road	704*576	774	66	IR(2886) IN(448) OV(924)	80
T2 Street	1600*1200	359	21	IR(322) IN(57) OV(38)	63
T3 Sidewalk	1600*1200	749	27	IR(416) IN(66) OV(218)	91
T4 Street	1280*720	7674	44	IR(732) IN(186) OV(952)	175
T5 Sidewalk	1280*720	1074	18	IR(244) IN(18) OV(43)	82
T6 Park	768*576	1449	9	IR(38) IN(9) OV(21)	111
T7 Indoor	384*288	1029	19	IR(184) IN(23) OV(135)	89
T8 Hall	320*240	1334	8	IR(44) IN(2) OV(10)	52
T9 Sidewalk	352*288	8029	38	IR(820) IN(99) OV(482)	257

Table 1: The nine surveillance video in our experiment. In the table, T_I is the number of input frames, N_o is the number of tubes, and T_S is the number of synopsis frames. (IR - irrelevant, IN - intersecting, OV - overlapping)

Operating System	Windows 8 Professional 64bit
Hardware	CPU: Intel Core i3 3.30-GHz Memory: 8-GB
Compiler	Microsoft Visual Studio 2008

Table 2: Details of the running environment

frames of synopsis to the original video $FR = T_S/T_I$, where T_S and T_I are the length of the synopsis video and that of the input video. The more the reduction of frames in synopsis, the smaller the FR .

2) *Frame compact rate (CR)*. The CR metric is used to judge if the foreground in synopsis video is rearranged closely, which is calculated by

$$CR = \frac{1}{w \cdot h \cdot T_S} \sum_{t=1}^{T_S} \sum_{x=1}^w \sum_{y=1}^h \{1 | \text{if } p(x, y, t) \in \text{foreground in } V_S\} \quad (8)$$

where $p(x, y, t)$ represents a pixel of the t -th frame in synopsis video, w and h is the width and height of synopsis frame, respectively. The higher the CR is, the more compact tubes the synopsis video has.

3) *Overlap ratio (OR)*. The overlap ratio (*OR*) indicates the collision degree of tubes, which is computed by

$$OR = \frac{1}{w \cdot h \cdot T_S} \sum_{t=1}^{T_S} \sum_{x=1}^w \sum_{y=1}^h \{1 | \text{if } p(x, y, t) \in \text{the collision foreground in } V_S\} \quad (9)$$

685 where the collision foreground means the overlapping area of two or more tubes in the image domain. The fewer collisions between tubes in synopsis, the smaller the *OR*.

5.1. Analysis of Parameters

As mentioned in Section 3.4, the parameter q is a key factor that can be adjusted to compromise between condensation and collision for the generated synopsis video. The statistical results in Section 3.4 provide reference for selection of q . Here, we discuss its influence on the performance of our method using the metrics such as the *FR*, *CR*, and *OR*. According to the results in Figure 8, the parameter q shall be larger than 3 for cars and 5 for pedestrians. Figure 695 10 shows the performance curve versus the value of q for nine videos in our dataset, that is $FR \sim q$, $CR \sim q$ and $OR \sim q$. As can be seen from Figure 10, although the values of the *FR*, *CR*, and *OR* are different for different videos, the performance curves show almost the same changing pattern: the *FR* value increases with q , while the *CR* and *OR* change in the inverse way. That is to say, the larger the parameter q , the less collisions in the synopsis, but the longer the synopsis and the less objects in one frame. Therefore, we left the parameter q adjustable for users to meet their specific preference. We simply set $q = 3$ in the following experiments for fair comparison with other methods, which will be described in Section 5.2.

705 Another parameter needs to be set is the overlapping threshold, which is used to differentiate the relationships of intersecting and overlapping. We set the overlapping threshold in an experimental way by conducting several experiments on different videos. Figure 11 shows the performance curve versus different values of overlapping threshold in [1,10] for nine videos in our dataset. As can be seen from Figure 11, the value of overlapping threshold doesn't make much 710

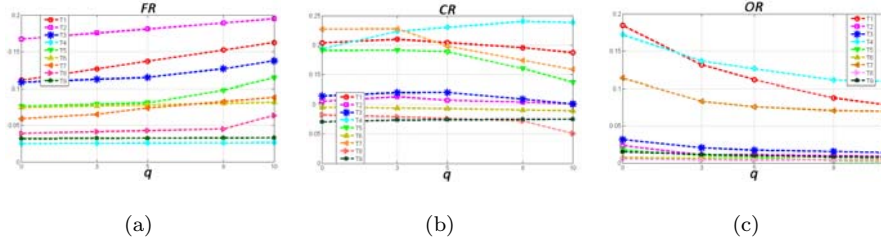


Figure 10: The performance of FR , CR , and OR influenced by parameter q .

difference to the results. Therefore, we set the overlapping threshold with an empiric value, which is 5 in the following experiments. The value is in line with intuition that two objects passing by each other in more than 5 frames are usually considered as overlapping.

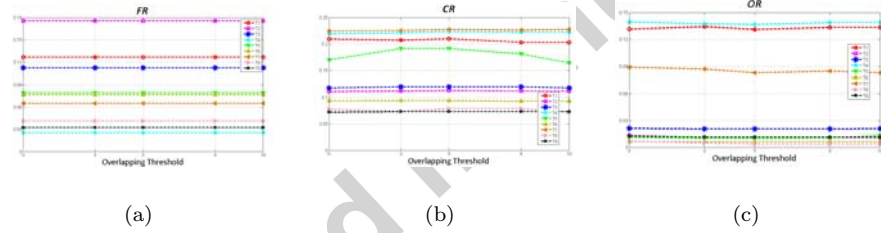
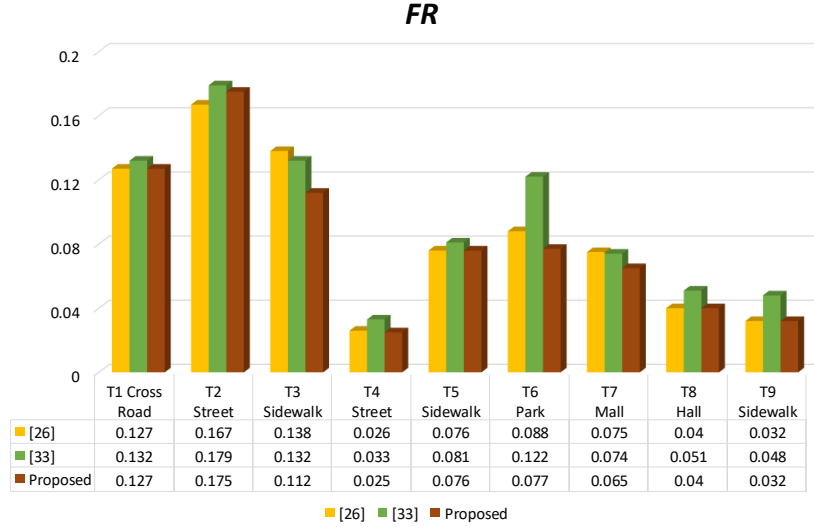


Figure 11: The performance of FR , CR , and OR influenced by overlapping threshold.

5.2. Performance Comparison

We compare our proposed method with those proposed by Y.Pritch [18], Huang[26], and Zhu[33]. The FR of [18] is not compared because it is manually set by users in [18]. Due to this reason, we set the same frame number in [18] as generated by our method in the comparison of the CR and OR . Figure 12-14 show the comparison results of the FR , CR and OR on the videos in our dataset. It can be seen from the comparison results that our proposed method achieves almost the best performance on the nine videos, that is, our method achieves lower FR and OR , but larger CR .

Next, we compare our method on different sequences with respect to the number of objects and their relationships. The proposed method achieves better

Figure 12: Comparison of *FR*

FR on sequences T4 and T9, while gives relative worse *FR* on sequences T2 and T1. The underlying reason is that the *FR* is related to the ratio of number of objects to the input frames, which can be seen from Table 1. For the *CR* and *OR*, the method gives better *CR* but worse *OR* on sequences T1, T4 and T7, while achieves worse *CR* but better *OR* on sequences T8 and T6. This is because the *CR* is directly proportional to the number of *IN* and *OV* (see Table 1), while *OR* is inversely. Analyzing the relationship between objects in T8 and T6 sequences, we find that there are less *IN* and *OV* relationship between tubes than other sequences, which are easier to deal with. But in the T1 and T4 sequences, where the number of overlapping relationships are more than the other sequences, making them difficult to be condensed in the timeline without collisions. The results exemplify the discussion in Section 3, which is the basis of our method. However, the proposed method of the *FR*, *CR*, *OR* on T1 and T4 sequences are much better than other methods. That is to say, our approach has the better ability on handling crowded video synopsis.

The illustrative images of the synopsis videos for T2, T3, T8, T9 are shown in Figure 15(a)-Figure 15(d), respectively. As shown in Figure 15(a), the man

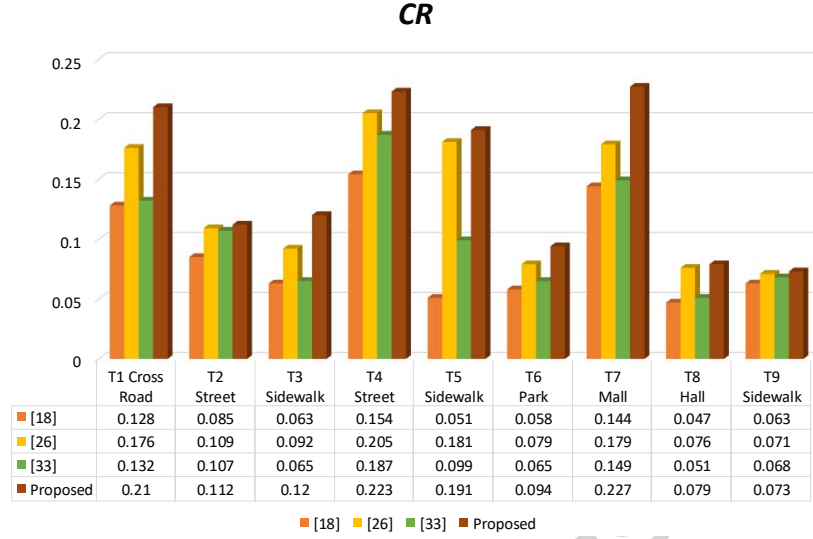


Figure 13: Comparison of CR

on bicycle, the silver car, and the man in red cloth which appear at different time of original video are detected and rearranged in the same frame of synopsis video. Similar results are presented in other scenes, which are street, hall, and sidewalk.

5.3. Complexity Analysis

In this subsection, we analyze the computational complexity of our proposed graph coloring based tube rearrangement. We compare the proposed approach to the optimization step in [18], which is the key contribution of this paper. Other steps in synopsis generation, such as background modeling, object detection and tracking are not considered here. The methods in [26, 33] are not compared here because they work in an online manner.

Due to that the condensation ratio, *i.e.* the number of frames, is manually assigned, we set the condensation ratio is the same as our proposed method for fair comparison of computational complexity. Figure 16 shows the running time for tube rearrangement of the two methods. For the optimization step in [18], temporal shifts in jumps of 5 frames and 10 frames are considered. For

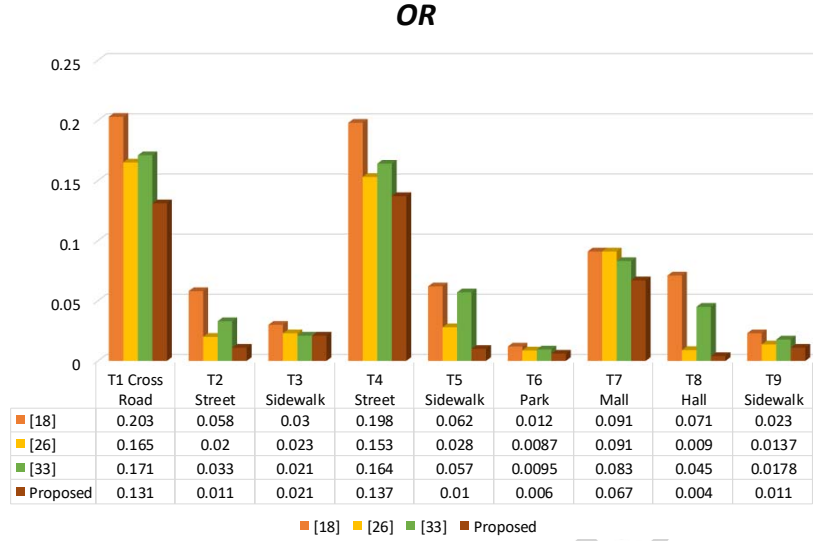


Figure 14: Comparison of *OR*

the proposed method, the time for tube rearrangement includes the time on
 760 relationship computation and the time on graph construction and coloring.

It can be seen from Figure 16 that, the proposed method has faster speed
 than the optimization step in [18], especially when the jump frames is smaller
 which means more iterations are needed to reach the final result. This is because
 the proposed method only computes the relationships between tubes only once,
 765 whereas the method in [18] computes the energy between tubes in each iteration
 of the optimization process, which is a huge redundancy.

Let K be numbers of tubes in the original video, N be the length of the syn-
 opsis and the computation burden of relationships between two tubes or pairwise
 energy function in [18] is denoted as CBT for convenience. The computation
 770 cost for tube rearrangement in [18] is $(K - 1) (N/k)^K \bullet CBT$, where k is the
 frame jump that can be adjusted by users for speed. And the computation cost
 for our method is approximately computed as $K (K - 1) \bullet CBT + O(N \bullet m^2)$,
 where m denotes the number of *s-nodes* in the potential collision graph, *i.e.* the
 number of potential collision points between tubes. That is to say, the com-
 775 plexity depends on the number of objects and the length of synopsis, which in-



Figure 15: Example synopsis results on different surveillance data sets.

creases exponentially, while the time complexity of the proposed method mainly depends on the relationships of tubes in the video. The more overlapping relations there are in original videos, the larger and connected the potential collision graph is, which leads to slower speed.

780 5.4. Discussion

1) *Relation to previous work.* In earlier methods such as [18, 26, 33], on-line or offline, all formulate the tube rearrangement by minimizing some certain global function and solve it through an optimization technique. Moreover, the computation of pairwise energy function is computed iteratively in the opti-

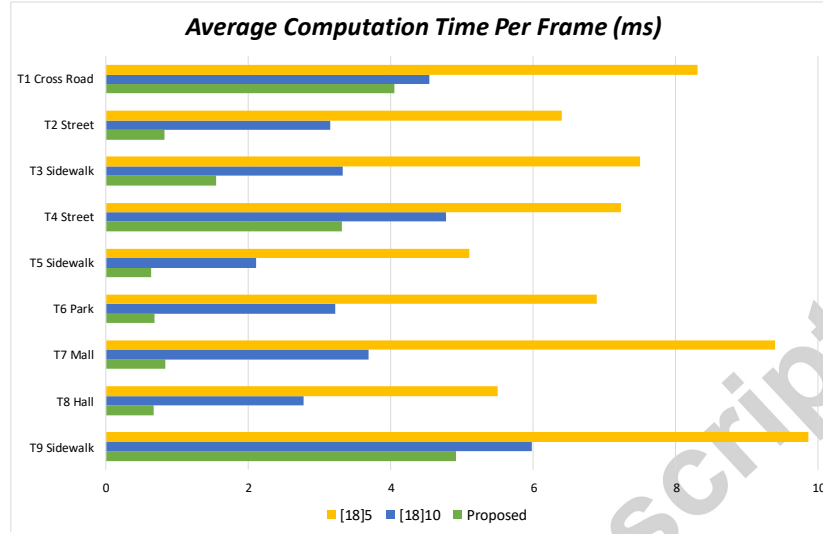


Figure 16: Comparison of running time.

785 mization process, which is a huge computation redundancy. In our paper, we analyze the reason behind collision phenomenon in essence and define three kinds of relationship for tubes in the original video. Based on that, we change the process flow of tube rearrangement by constructing potential collision graph to representing the relationships between tubes and formulating it as a graph coloring problem. In this way, the repetitive computation of pairwise relationship is saved and the tube rearrangement can be solved perfectly using graph coloring algorithms. Besides, the length of the synopsis in earlier work is not an optimized value, either manually set beforehand [18] or randomly determined [26, 33]. But in the proposed method it is also optimized because the PCG graph is colored using as less colors as possible, *i.e.* as shorter as possible for synopsis videos.

2) *Limitations and Future research.* In the whole flow of synopsis generation, several steps such as background modeling, object detection and tracking, and object stitching are required. However, we mainly focus on the tube rearrangement step in this paper and other steps are only described briefly. But they are important as well and affect the performance of the proposed tube rear-

rangement to some extent. Development on those steps will definitely improve
 the quality of the final synopsis. Moreover, several factors need to be taken
 into consideration to generate a synopsis according to different requirements of
 805 users, such as the chronological order, motion structure, activity preserving and
 so on, but we mainly emphasize the collision issue here and leave other factors
 for future research. The collision eliminating is solved perfectly by our graph
 coloring framework, and future work will taking other factors into consideration
 by imposing more constraints on $L(q)$ -coloring of the graph. A future work is
 810 adaptively setting parameter q for better condensation and less collisions. In
 this paper, we set the value of q statistically by counting the frames objects pass
 by a pixel in the image, which will achieve a median performance with respect
 to condensation and collision. In the future, we will try to give the parameter
 q adaptive values by weighting the edges between two tubes according to their
 815 size and speed.

6. Conclusion

In this paper, we proposed a novel approach for tube rearrangement in video
 synopsis. Unlike conventional methods, our method solves tube rearrangement
 in video synopsis by formulating it as a graph coloring problem. In our ap-
 820 proach, the original video is represented by a graph of main nodes (m -node)
 composed of sub-nodes(s -node), each m -node of which stands for a tube from
 the video and each s -node on an m -node indicates a potential collision point
 between the tube with another. Therefore the edges between s -nodes describe
 the potential collision relationship between events. To mitigate the collision
 825 artifacts, our method finds the mapping of events from the original video to
 the synopsis video by $L(q)$ -coloring the graph, which arranges the tube in the
 synopsis by positioning the collision time instead of the starting time. The
 parameter q is left adjustable to make a compromise between collision artifacts
 and synopsis length to better meet user's demand. Then, the shifted objects are
 830 finally composited with the background image to obtain the high-quality video

synopsis. Extensive experimental results have shown that the proposed method can generate more compact synopsis video free of collision artifacts with less computational complexity.

References

- 835 [1] D. O'callaghan, E. L. Lew, Method and apparatus for video on demand with fast forward, reverse and channel pause, uS Patent 5,477,263 (Dec. 19 1995).
- [2] N. Petrovic, N. Jojic, T. S. Huang, Adaptive video fast forward, *Multimedia Tools and Applications* 26 (3) (2005) 327–344.
- 840 [3] B. Höferlin, M. Höferlin, D. Weiskopf, G. Heidemann, Information-based adaptive fast-forward for visual surveillance, *Multimedia Tools and Applications* 55 (1) (2011) 127–150.
- [4] M. Smith, T. Kanade, et al., Video skimming and characterization through the combination of image and language understanding, in: *Content-Based Access of Image and Video Database, 1998. Proceedings., 1998 IEEE International Workshop on*, IEEE, 1998, pp. 61–70.
- 845 [5] C. Kim, J.-N. Hwang, An integrated scheme for object-based video abstraction, in: *Proceedings of the eighth ACM international conference on Multimedia*, ACM, 2000, pp. 303–311.
- 850 [6] J. Oh, Q. Wen, S. Hwang, J. Lee, Video abstraction, *Video data management and information retrieval* (2004) 321–346.
- [7] Y. Li, T. Zhang, D. Tretter, An overview of video abstraction techniques, Tech. rep., Technical Report HPL-2001-191, HP Laboratory (2001).
- [8] M. M. Yeung, B.-L. Yeo, Video visualization for compact presentation and fast browsing of pictorial content, *Circuits and Systems for Video Technology*, IEEE Transactions on 7 (5) (1997) 771–785.
- 855

- [9] X. Zhu, X. Wu, J. Fan, A. K. Elmagarmid, W. G. Aref, Exploring video content structure for hierarchical summarization, *Multimedia Systems* 10 (2) (2004) 98–115.
- 860 [10] Y.-F. Ma, L. Lu, H.-J. Zhang, M. Li, A user attention model for video summarization, in: *Proceedings of the tenth ACM international conference on Multimedia*, ACM, 2002, pp. 533–542.
- [11] N. Ejaz, T. B. Tariq, S. W. Baik, Adaptive key frame extraction for video summarization using an aggregation mechanism, *Journal of Visual Communication and Image Representation* 23 (7) (2012) 1031–1040.
- 865 [12] J. Niu, D. Huo, K. Wang, C. Tong, Real-time generation of personalized home video summaries on mobile devices, *Neurocomputing* 120 (2013) 404–414.
- [13] X. Song, L. Sun, J. Lei, D. Tao, G. Yuan, M. Song, Event-based large scale surveillance video summarization, *Neurocomputing* 187 (2016) 66–74.
- 870 [14] L. dos Santos Belo, C. A. Caetano, Z. K. G. do Patrocínio, S. J. F. Guimarães, Summarizing video sequence using a graph-based hierarchical approach, *Neurocomputing* 173 (2016) 1001–1016.
- [15] H.-W. Kang, Y. Matsushita, X. Tang, X.-Q. Chen, Space-time video montage, in: *computer vision and pattern recognition, 2006 IEEE computer society conference on*, Vol. 2, IEEE, 2006, pp. 1331–1338.
- 875 [16] M. Irani, P. a. Anandan, J. Bergen, R. Kumar, S. Hsu, Efficient representations of video sequences and their applications, *Signal Processing: Image Communication* 8 (4) (1996) 327–351.
- 880 [17] C. Pal, N. Jojic, Interactive montages of sprites for indexing and summarizing security video, in: *Computer Vision and Pattern Recognition, 2005. CVPR 2005. IEEE Computer Society Conference on*, Vol. 2, IEEE, 2005, pp. 1192–vol.

- [18] Y. Pritch, A. Rav-Acha, S. Peleg, Nonchronological video synopsis and indexing, *Pattern Analysis and Machine Intelligence*, IEEE Transactions on 30 (11) (2008) 1971–1984.
- [19] Y. Pritch, A. Rav-Acha, A. Gutman, S. Peleg, Webcam synopsis: Peeking around the world, in: *Computer Vision, 2007. ICCV 2007. IEEE 11th International Conference on*, IEEE, 2007, pp. 1–8.
- [20] A. Rav-Acha, Y. Pritch, S. Peleg, Making a long video short: Dynamic video synopsis, in: *Computer Vision and Pattern Recognition, 2006 IEEE Computer Society Conference on*, Vol. 1, IEEE, 2006, pp. 435–441.
- [21] W. Lin, Y. Zhang, J. Lu, B. Zhou, J. Wang, Y. Zhou, Summarizing surveillance videos with local-patch-learning-based abnormality detection, blob sequence optimization, and type-based synopsis, *Neurocomputing* 155 (2015) 84–98.
- [22] Y. Pritch, S. Ratovitch, A. Hende, S. Peleg, Clustered synopsis of surveillance video, in: *Advanced Video and Signal Based Surveillance, 2009. AVSS'09. Sixth IEEE International Conference on*, IEEE, 2009, pp. 195–200.
- [23] M. Xu, S. Z. Li, B. Li, X.-T. Yuan, S.-M. Xiang, A set theoretical method for video synopsis, in: *Proceedings of the 1st ACM international conference on Multimedia information retrieval*, ACM, 2008, pp. 366–370.
- [24] S. Kasamwattananarote, N. Cooharajanane, S. Satoh, R. Lipikorn, Real time tunnel based video summarization using direct shift collision detection, in: *Advances in Multimedia Information Processing-PCM 2010*, Springer, 2010, pp. 136–147.
- [25] C.-R. Huang, H.-C. Chen, P.-C. Chung, Online surveillance video synopsis, in: *Circuits and Systems (ISCAS), 2012 IEEE International Symposium on*, IEEE, 2012, pp. 1843–1846.

- [26] C.-R. Huang, P.-C. J. Chung, D.-K. Yang, H.-C. Chen, G.-J. Huang, Maximum a posteriori probability estimation for online surveillance video synopsis, *Circuits and Systems for Video Technology*, IEEE Transactions on 24 (8) (2014) 1417–1429.
- 915 [27] Y. Nie, C. Xiao, H. Sun, P. Li, Compact video synopsis via global spatiotemporal optimization, *Visualization and Computer Graphics*, IEEE Transactions on 19 (10) (2013) 1664–1676.
- [28] L. Sun, J. Xing, H. Ai, S. Lao, A tracking based fast online complete video synopsis approach, in: *Pattern Recognition (ICPR)*, 2012 21st International Conference on, IEEE, 2012, pp. 1956–1959.
- 920 [29] M. Lu, Y. Wang, G. Pan, Generating fluent tubes in video synopsis, in: *Acoustics, Speech and Signal Processing (ICASSP)*, 2013 IEEE International Conference on, IEEE, 2013, pp. 2292–2296.
- [30] R. Zhong, R. Hu, Z. Wang, S. Wang, Fast synopsis for moving objects using compressed video, *Signal Processing Letters*, IEEE 21 (7) (2014) 834–838.
- 925 [31] W. Fu, J. Wang, L. Gui, H. Lu, S. Ma, Online video synopsis of structured motion, *Neurocomputing* 135 (2014) 155–162.
- [32] S. Feng, Z. Lei, D. Yi, S. Z. Li, Online content-aware video condensation, in: *Computer Vision and Pattern Recognition (CVPR)*, 2012 IEEE Conference on, IEEE, 2012, pp. 2082–2087.
- 930 [33] J. Zhu, S. Feng, D. Yi, S. Liao, Z. Lei, S. Z. Li, High-performance video condensation system, *IEEE Transactions on Circuits and Systems for Video Technology* 25 (7) (2015) 1113–1124.
- [34] X. Zhu, J. Liu, J. Wang, H. Lu, Key observation selection-based effective video synopsis for camera network, *Machine vision and applications* 25 (1) 935 (2014) 145–157.

- [35] J. Zhu, S. Liao, S. Z. Li, Multicamera joint video synopsis, *IEEE Transactions on Circuits and Systems for Video Technology* 26 (6) (2016) 1058–1069.
- 940 [36] J. R. Griggs, R. K. Yeh, Labelling graphs with a condition at distance 2, *SIAM Journal on Discrete Mathematics* 5 (4) (1992) 586–595.
- [37] M. R. Garey, D. S. Johnson, *Computers and intractability: a guide to the theory of np-completeness*. 1979, San Francisco, LA: Freeman.
- [38] D. Wood, A technique for colouring a graph applicable to large scale
945 timetabling problems, *The Computer Journal* 12 (4) (1969) 317–319.
- [39] D. De Werra, The combinatorics of timetabling, *European Journal of Operational Research* 96 (3) (1997) 504–513.
- [40] A. Mishra, S. Banerjee, W. Arbaugh, Weighted coloring based channel
950 assignment for wlans, *ACM SIGMOBILE Mobile Computing and Communications Review* 9 (3) (2005) 19–31.
- [41] V. Maan, G. Purohit, A novel frequency allocation mechanism with separation in mobile cellular network based on distributed setting, in: *Advanced Electronic Systems (ICAES), 2013 International Conference on*, IEEE, 2013, pp. 319–321.
- 955 [42] E. Karami, S. Glisic, Joint optimization of scheduling and routing in multicast wireless ad hoc networks using soft graph coloring and nonlinear cubic games, *Vehicular Technology, IEEE Transactions on* 60 (7) (2011) 3350–3360.
- [43] L. M. Abdullah, M. D. Baba, S. G. A. Ali, A. O. Lim, Y. Tan, New graph
960 colouring algorithm for resource allocation in large-scale wireless networks, in: *Control and System Graduate Research Colloquium (ICSGRC), 2014 IEEE 5th*, IEEE, 2014, pp. 233–238.

- [44] W. K. Hale, Frequency assignment: Theory and applications, *Proceedings of the IEEE* 68 (12) (1980) 1497–1514.
- 965 [45] S. Khor, Application of graph colouring to biological networks, *Systems Biology, IET* 4 (3) (2010) 185–192.
- [46] M. Gendreau, G. Laporte, F. Semet, Heuristics and lower bounds for the bin packing problem with conflicts, *Computers & Operations Research* 31 (3) (2004) 347–358.
- 970 [47] Y. Ye, L. Yi-jun, W. Yan-qing, An improved aco algorithm for the bin packing problem with conflicts based on graph coloring model, in: *Management Science & Engineering (ICMSE), 2014 International Conference on*, IEEE, 2014, pp. 3–9.
- [48] E. Driouch, W. Ajib, On the user scheduling in cognitive radio mimo networks, in: *Global Communications Conference (GLOBECOM), 2012 IEEE*, IEEE, 2012, pp. 1242–1247.
- 975 [49] F. S. Roberts, *Graph theory and its applications to problems of society*, SIAM, 1978.
- [50] W. Klotz, Graph coloring algorithms, *Mathematics Report* 5 (2002) (2002) 1–9.
- 980 [51] E. G. Boman, D. Bozdağ, U. Catalyurek, A. H. Gebremedhin, F. Manne, A scalable parallel graph coloring algorithm for distributed memory computers, in: *Euro-Par 2005 Parallel Processing*, Springer, 2005, pp. 241–251.
- [52] D. Bozdağ, U. Catalyurek, A. H. Gebremedhin, F. Manne, E. G. Boman, F. Özgüner, A parallel distance-2 graph coloring algorithm for distributed memory computers, in: *High Performance Computing and Communications*, Springer, 2005, pp. 796–806.
- 985 [53] S. Sengupta, Parallel graph coloring algorithms on the gpu using opencl, in: *Computing for Sustainable Global Development (INDIACom), 2014 International Conference on*, IEEE, 2014, pp. 353–357.
- 990

- [54] N. M. Gandhi, R. Misra, Performance comparison of parallel graph coloring algorithms on bsp model using hadoop, in: Computing, Networking and Communications (ICNC), 2015 International Conference on, IEEE, 2015, pp. 110–116.
- 995 [55] M.-H. Tayarani-N, A. Prügel-Bennett, Anatomy of the fitness landscape for dense graph-colouring problem, *Swarm and Evolutionary Computation* 22 (2015) 47–65.
- [56] M.-H. Tayarani-N, A. Prugel-Bennett, On the landscape of combinatorial optimization problems, *Evolutionary Computation*, *IEEE Transactions on* 18 (3) (2014) 420–434.
- 1000 [57] P. Galinier, A. Hertz, A survey of local search methods for graph coloring, *Computers & Operations Research* 33 (9) (2006) 2547–2562.
- [58] M. Baghel, S. Agrawal, S. Silakari, Recent trends and developments in graph coloring, in: *Proceedings of the International Conference on Frontiers of Intelligent Computing: Theory and Applications (FICTA)*, Springer, 2013, pp. 431–439.
- 1005 [59] J. Kwon, K. M. Lee, A unified framework for event summarization and rare event detection from multiple views, *IEEE transactions on pattern analysis and machine intelligence* 37 (9) (2015) 1737–1750.
- 1010 [60] D. Brélaz, New methods to color the vertices of a graph, *Commun. ACM* 22 (4) (1979) 251–256.
- [61] F. T. Leighton, A graph coloring algorithm for large scheduling problems, *Journal of research of the national bureau of standards* 84 (6) (1979) 489–506.
- 1015 [62] O. Barnich, M. Van Droogenbroeck, Vibe: A universal background subtraction algorithm for video sequences, *Image Processing, IEEE Transactions on* 20 (6) (2011) 1709–1724.

- 1020 [63] K. Choeychuen, P. Kumhom, K. Chamnongthai, An efficient implementation of the nearest neighbor based visual objects tracking, in: Intelligent Signal Processing and Communications, 2006. ISPACS'06. International Symposium on, IEEE, 2006, pp. 574–577.
- [64] P. Pérez, M. Gangnet, A. Blake, Poisson image editing, in: ACM Transactions on Graphics (TOG), Vol. 22, ACM, 2003, pp. 313–318.